

Failure Modes of LLM Systems: Why They Break Even When They “Work”

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Lecture 5

Convincing Failure

The most dangerous failures are those that appear correct.

- ▶ Outputs are fluent and well-structured.
- ▶ Errors are embedded within plausible explanations.
- ▶ Confidence is simulated rather than earned.
- ▶ Users tend to trust coherent language.
- ▶ Failures require scrutiny to detect.
- ▶ The system fails in ways that resemble success.

Bridge from Phase 2

Structured workflows improve reliability but do not eliminate failure.

- ▶ Scaffolding improves reliability but not correctness.
- ▶ Transformation pipelines reduce risk but do not remove it.
- ▶ Role separation limits but does not stop error propagation.
- ▶ Systems remain probabilistic.
- ▶ Failures persist under complexity.
- ▶ Reliability remains conditional.

Taxonomy of Failure

Failures can be categorized into distinct types.

- ▶ Epistemic failures concern truth.
- ▶ Structural failures concern consistency.
- ▶ Process failures concern interaction.
- ▶ Each type requires different mitigation.
- ▶ Failures often overlap categories.
- ▶ Taxonomy enables diagnosis.

Epistemic Failures

Epistemic failures produce plausible but incorrect outputs.

- ▶ Statements are not grounded in data.
- ▶ Outputs appear analytical.
- ▶ Confidence is unrelated to correctness.
- ▶ Errors require verification to detect.
- ▶ Coherence masks inaccuracy.
- ▶ Truth is not internally validated.

Example: Fabricated Analysis

Models generate convincing but incorrect analytical conclusions.

- ▶ Trends inferred from insufficient data.
- ▶ Quantitative claims invented.
- ▶ Statistical reasoning simulated.
- ▶ Outputs resemble valid analysis.
- ▶ No numerical verification occurs.
- ▶ External validation is required.

Simulation vs Analysis

The model simulates reasoning rather than performing it.

- ▶ Pattern matching replaces computation.
- ▶ Language structure substitutes logic.
- ▶ Outputs mimic analytical formats.
- ▶ No execution of statistical methods.
- ▶ Results reflect training patterns.
- ▶ Plausibility replaces correctness.

Implications of Epistemic Failure

Epistemic failures are dangerous in analytical contexts.

- ▶ Research conclusions may be invalid.
- ▶ Decisions may be misled.
- ▶ Incorrect insights appear authoritative.
- ▶ Novices are especially vulnerable.
- ▶ Errors scale with trust.
- ▶ Verification becomes mandatory.

Fabrication and Hallucination

Models generate content that appears valid but is invented.

- ▶ Citations may not exist.
- ▶ References may be misattributed.
- ▶ Test outputs may be fabricated.
- ▶ Explanations lack grounding.
- ▶ Content appears complete.
- ▶ Uncertainty is not indicated.

Nature of Hallucination

Hallucinations follow predictable patterns.

- ▶ Occur when information is missing.
- ▶ Gaps are filled with patterns.
- ▶ Outputs reflect likelihood, not truth.
- ▶ Fluency increases credibility.
- ▶ Errors are systematic.
- ▶ Detection requires validation.

Structural Failures

Structural failures arise from inconsistency across components.

- ▶ Schema and code may misalign.
- ▶ Representations may diverge.
- ▶ Shared structure may be lost.
- ▶ Errors propagate through stages.
- ▶ Systems become incoherent.
- ▶ Correctness breaks across layers.

Example: Representation Drift

Complex systems reveal structural instability.

- ▶ JSON differs from implementation.
- ▶ Code violates defined structure.
- ▶ Tests fail from mismatches.
- ▶ Changes do not propagate.
- ▶ Systems become brittle.
- ▶ Alignment is lost over time.

Lack of Internal Consistency

Models do not enforce consistency across outputs.

- ▶ Each output is independent.
- ▶ No persistent internal model exists.
- ▶ Prior structure is not preserved.
- ▶ Alignment must be enforced externally.
- ▶ Consistency degrades without constraints.
- ▶ Structure is not automatic.

Process Failures

Process failures arise from the interaction itself.

- ▶ Conversations become unstable.
- ▶ Outputs drift from intent.
- ▶ Irrelevant content appears.
- ▶ Style degrades over time.
- ▶ Context becomes noisy.
- ▶ Workflow loses coherence.

Example: Context Drift

Extended interaction introduces instability.

- ▶ Instructions are forgotten.
- ▶ Irrelevant artifacts appear.
- ▶ Responses lose focus.
- ▶ Format becomes inconsistent.
- ▶ Errors influence future outputs.
- ▶ Conversation diverges.

Verbosity and Degradation

Output quality changes over time without control.

- ▶ Responses become longer.
- ▶ Precision decreases.
- ▶ Explanations expand unnecessarily.
- ▶ Formatting becomes inconsistent.
- ▶ Signal-to-noise decreases.
- ▶ User must constrain output.

Interaction Instability

The interaction itself introduces error.

- ▶ Responses condition future responses.
- ▶ Errors accumulate.
- ▶ Context mixes correct and incorrect data.
- ▶ Model cannot distinguish them.
- ▶ Feedback loops reinforce mistakes.
- ▶ Stability must be managed.

Silent Degradation

Errors accumulate without immediate detection.

- ▶ Early inaccuracies go unnoticed.
- ▶ Model reuses incorrect outputs.
- ▶ Errors compound.
- ▶ System appears consistent.
- ▶ Detection becomes difficult.
- ▶ Failures emerge late.

User Silence as Reinforcement

Lack of correction reinforces errors.

- ▶ Incorrect outputs are accepted.
- ▶ Model adapts to perceived approval.
- ▶ Errors persist in future outputs.
- ▶ Mistakes become entrenched.
- ▶ Oversight is required.
- ▶ Correction must be explicit.

Why Failures Occur

Failures derive from fundamental system limitations.

- ▶ No internal model of truth exists.
- ▶ No persistent state is maintained.
- ▶ No verification mechanism is present.
- ▶ Outputs are generated token-by-token.
- ▶ Correctness is not evaluated.
- ▶ Consistency is not enforced.

Core Conclusion

Reliability must be imposed rather than assumed.

- ▶ Structure reduces error.
- ▶ Verification must be external.
- ▶ Constraints must be explicit.
- ▶ Oversight must be continuous.
- ▶ Systems must anticipate failure.
- ▶ Trust requires validation.

Practical Implications

Effective use requires disciplined control.

- ▶ Verify outputs against sources.
- ▶ Use structured inputs.
- ▶ Restart conversations when needed.
- ▶ Limit context to relevance.
- ▶ Treat outputs as drafts.
- ▶ Do not trust fluency.

Operational Rules

A set of rules mitigates common failures.

- ▶ Ground work in real data.
- ▶ Constrain outputs explicitly.
- ▶ Separate roles and tasks.
- ▶ Validate at each stage.
- ▶ Detect errors early.
- ▶ Maintain workflow control.

Synthesis

Failures are predictable and systematic.

- ▶ Epistemic failures distort truth.
- ▶ Structural failures break systems.
- ▶ Process failures destabilize workflows.
- ▶ All are inherent to the system.
- ▶ None are eliminated by scale.
- ▶ All require external control.

Transition to Lecture 6

Failures extend beyond individual use into larger systems.

- ▶ Writing errors become misinformation.
- ▶ Code errors become system failures.
- ▶ Analysis errors become decisions.
- ▶ Risks increase at scale.
- ▶ Systems amplify failure impact.
- ▶ Next topic is security and governance.

References I